

UDROŹNIENIE ŁÓDZKIEGO WĘZŁA KOLEJOWEGO (TEN-T), ETAP II, ODCINEK ŁÓDŹ FABRYCZNA - ŁÓDŹ KALISKA/ŁÓDŹ ŻABIENIEC – PROJEKT POIIS 5.1-15

TBM 13.08 M - CONSTRUCTION FOLLOW-UP

IDENTIFICATION TABLE

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1. INTRODUCTION

S-1222 TBM EPB machine with 13.08 m excavation diameter is currently mining Axis 23, in the section between Polesie Station and Śródmieście Station. TBM has restarted from chainage 1+962.

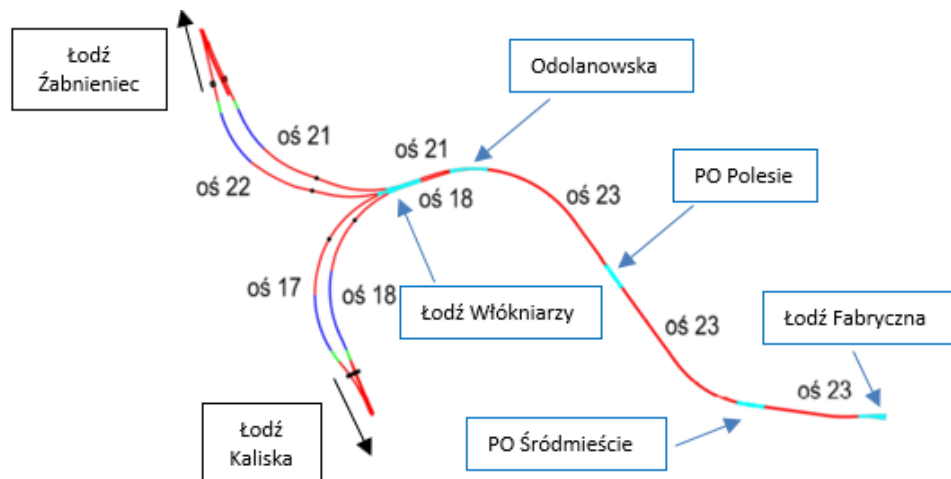


Figure 1-1 Overview of main structural elements of the Project

This daily follow-up report aims to give a general overview of S-1222 TBM drive during 30/08/2024-01/09/2024. It includes EPB face pressure, belt scale parameters, thrust and torque values, primary grout parameters, guidance, as well as monitoring follow-up data.

According to VMT system, TBM is at Chainage 1+656.84 after the erection of ring n. 251; which corresponds to a total excavation length of 309.63 m from the starting chainage.

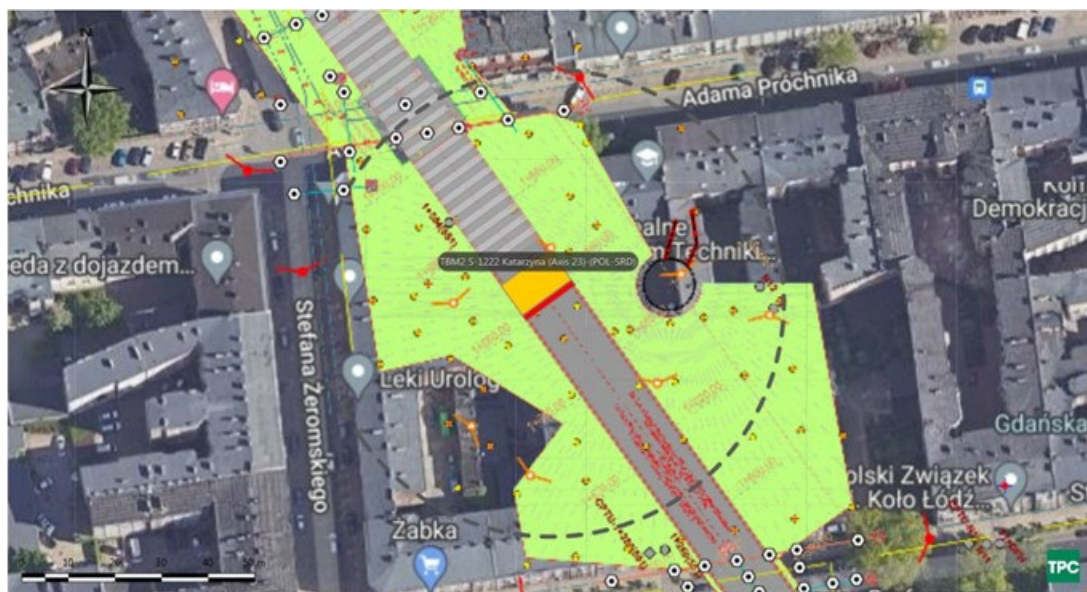


Figure 1-2 Plan view of TBM Position on 02/09/2024

2. FACE PRESSURES AND EXCAVATION VOLUMES

Average EPB pressure at the crown is at about 1.7 bar. The value is within the attention values.

There are drops of face pressure during the excavation of ring 245, 246, 247, 249, with values that range between 1.2 bar and 0.8 bar, lower than alarm thresholds. The pressure has then returned within attention limits.

Figure 2-1 shows EPB sensors.

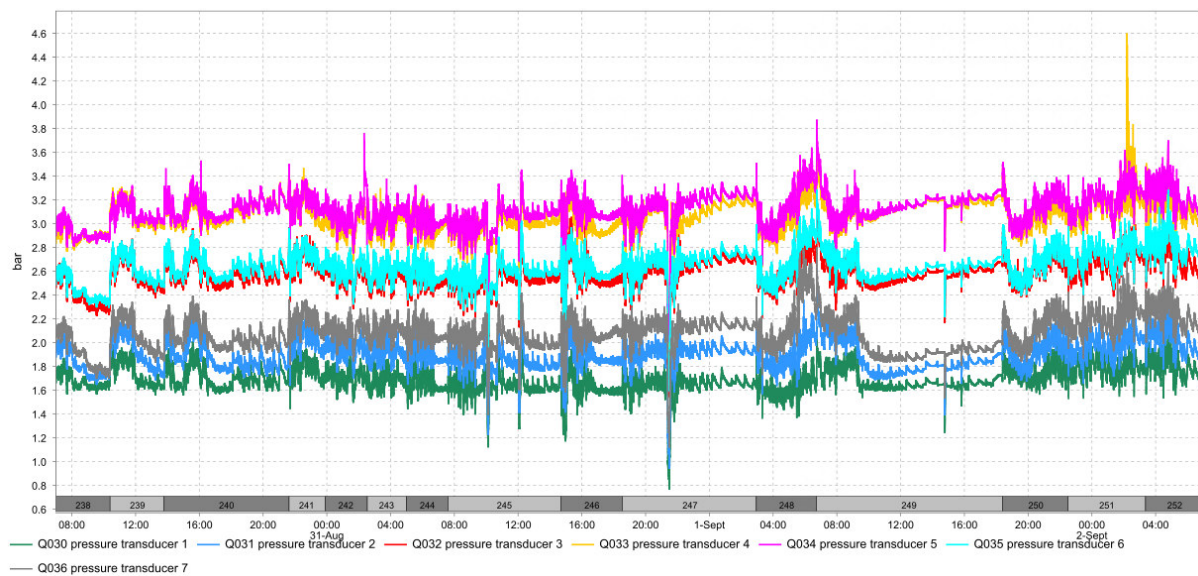


Figure 2-1 EPB pressure sensor

The delta between pressure transducer 1 and 2 has an average value of 0.22 bar indicating that the bulk chamber is correctly filled with material. At the end of ring 238, 239 and 240 and during excavation of rings 245, 249 and 250, drops of delta has occurred, until values lower than 0.1 bar indicating that the upper part of the chamber is not correctly filled with conditioned soil.

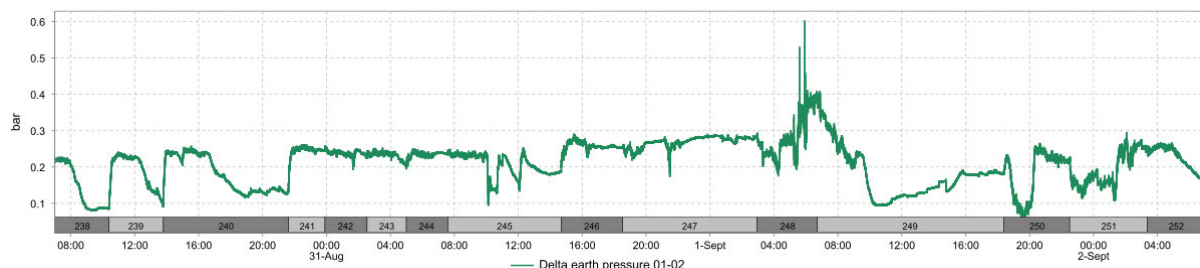


Figure 2-2 Delta between pressure sensors 1 and 2

Belt scale n. 2 show values much lower than n. 1. The results of Belt scale n. 2 seems to be affected by a constant error and are therefore considered not representative. The belt scale value is a raw indication of the excavated material quantity.

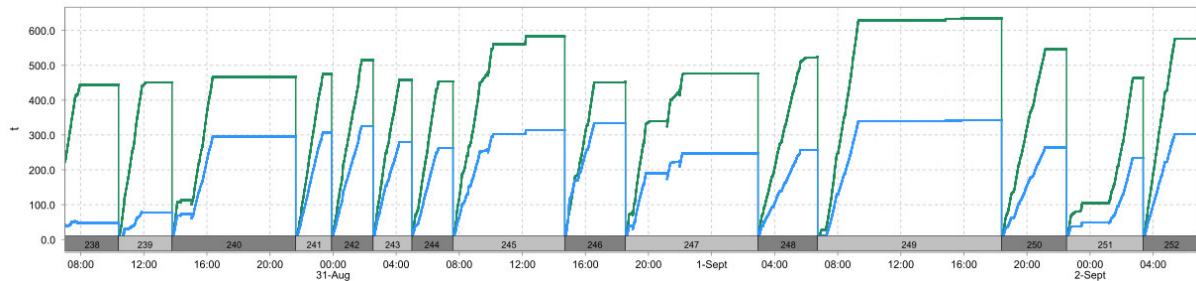


Figure 2-3 Belt scales

Table 2-1 Belt scale values at the end of each advance

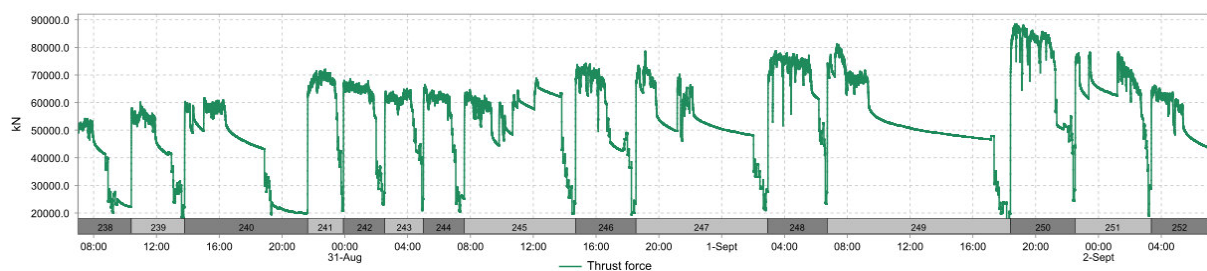
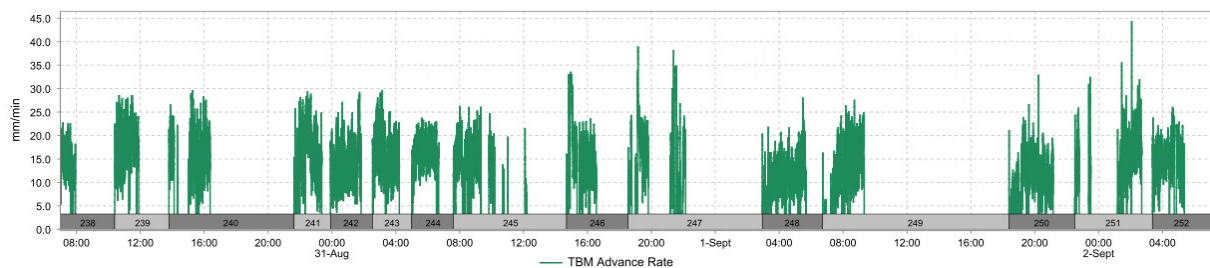
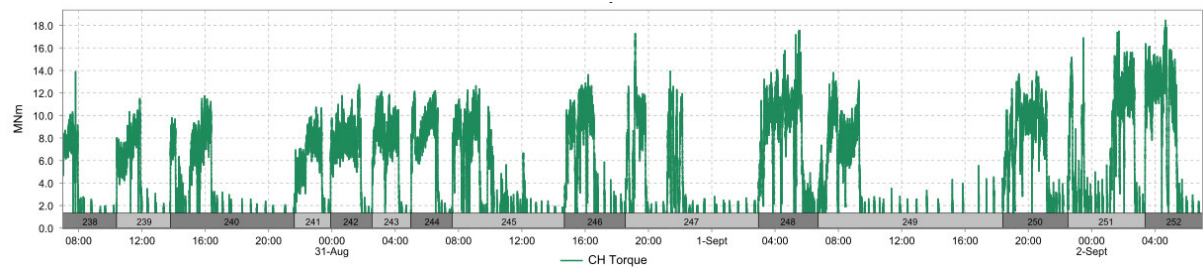
Ring	Belt scale 1 (t)	Belt scale 2 (t)
238	444	53
239	451	79
240	467	296
241	475	307
242	515	325
243	458	280
244	454	262
245	583	315
246	452	333
247	477	247
248	524	257
249	635	342
250	547	264
251	465	234
252	577	302

3. THRUST AND CUTTERHEAD TORQUE

During excavation, the average torque is of 8.7 MNm.

During advancement, average advance rate is of 15 mm/min.

During excavation, thrust force shows an average value of 65.2 MN and a maximum of 87.6 MN.



4. GROUT INJECTION

Grout injection has a volume above the minimum theoretical annular volume $+10\%$ of 13.54 m^3 , except for rings 238, 241, 246 and 252 that have lower volumes, much closer to theoretical value. Ratio B/A is at 7% .

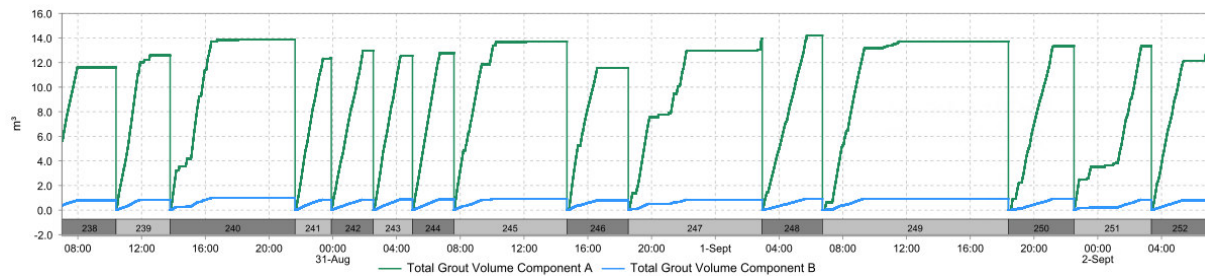


Figure 4-1 Grout volumes

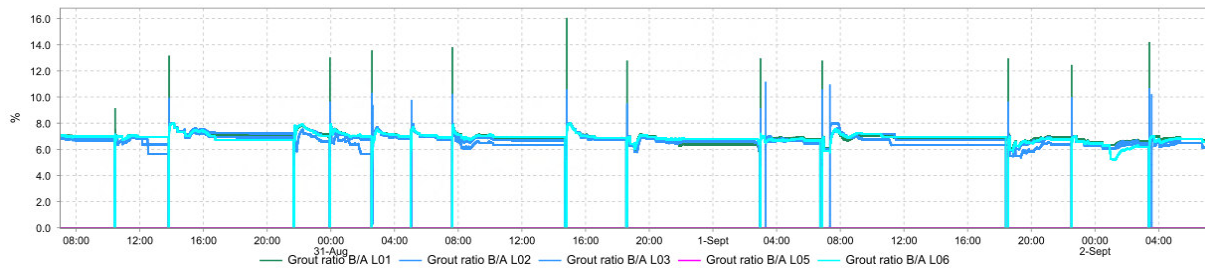


Figure 4-2 Volume ratio Components B/A

Table 4-1 Grout volumes per ring

Ring	Component A (m^3)	Component B (m^3)	B/A (%)
238	11.6	0.8	7
239	12.6	0.8	7
240	13.9	1	7
241	12.4	0.9	7
242	12.9	0.9	7
243	12.5	0.9	7
244	12.8	0.9	7
245	13.7	0.9	7
246	11.6	0.8	7
247	13.9	0.9	6

Ring	Component A (m³)	Component B (m³)	B/A (%)
248	14.2	0.9	7
249	13.7	0.9	7
250	13.3	0.9	7
251	13.3	0.9	6
252	12.7	0.8	6

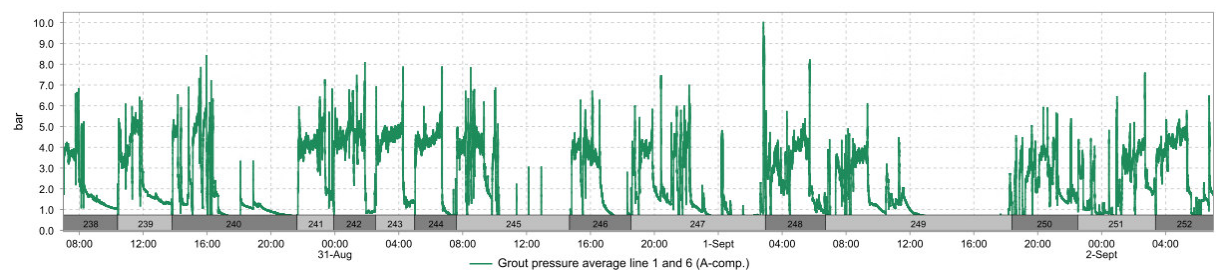


Figure 4-3 Average injection pressure line 1 and 6

5. GUIDANCE

The TBM vertical deviation is of about -3.46 mm on the front edge of the shield and the horizontal deviation of about -20.53 mm (Figure 5-1). Maximum radial deviation is of 40.54 mm.

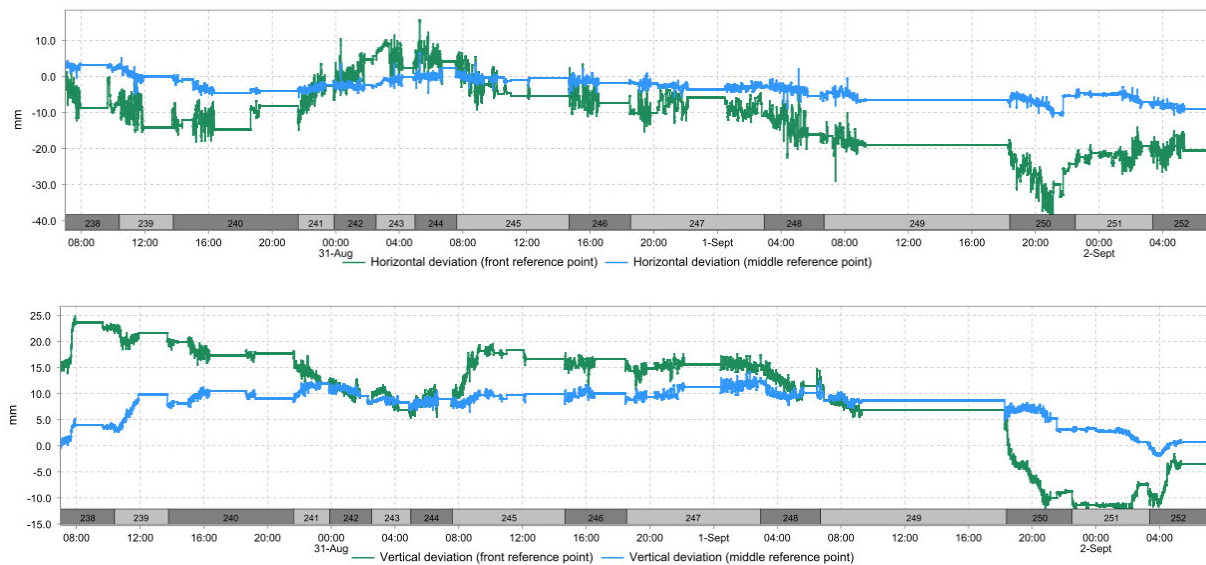


Figure 5-1 Machine deviation from the alignment

6. MONITORING

According to Geoscope, settlements of the buildings above the tunnel are kept within the limits.



Figure 6-1 Monitoring points

The monitoring points on Prochnika street are stable.

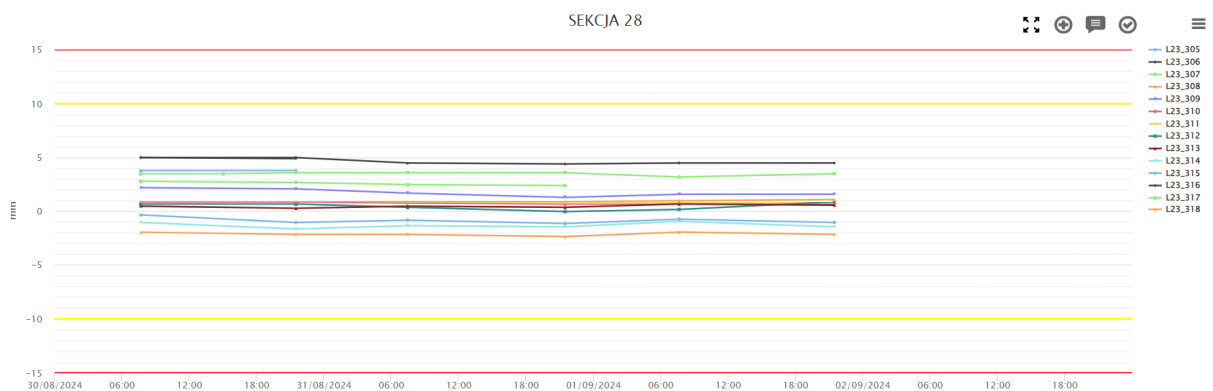


Figure 6-2 Settlement of Section on Prochnika street

On PR0430, monitoring points are stable.

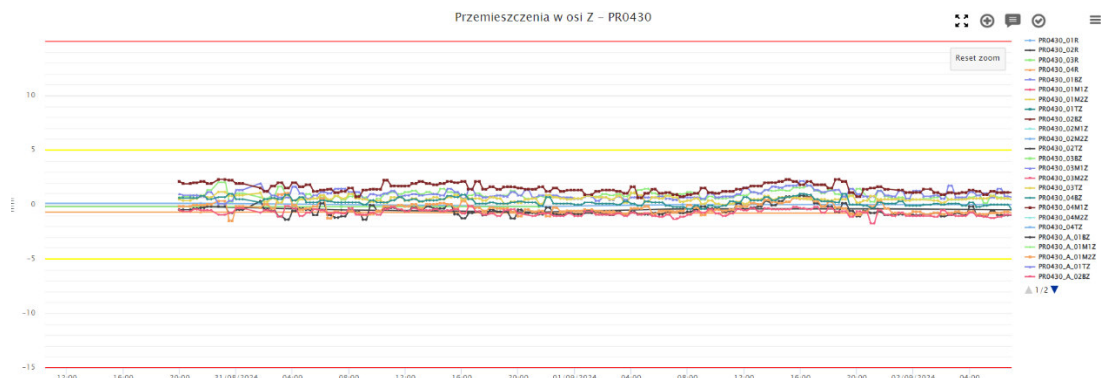


Figure 6-3 Settlement of building PR0430

On PR0440, monitoring points show a downward trend for points PR0440-03R, PR0440-04R, PR0440-05R, closer to tunnel axis, with a maximum settlement of about 12 mm in 72 hours, higher than the attention threshold.

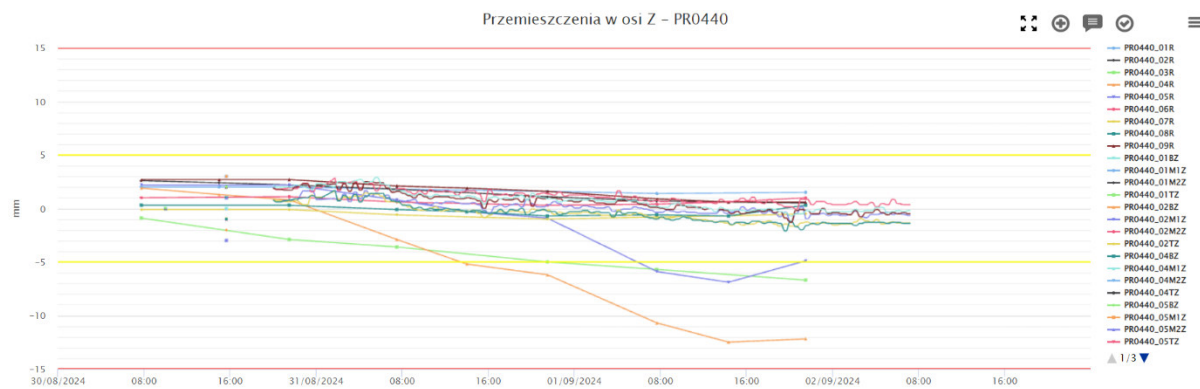


Figure 6-4 Settlement of building PR0440

On PR0450, monitoring points are stable.

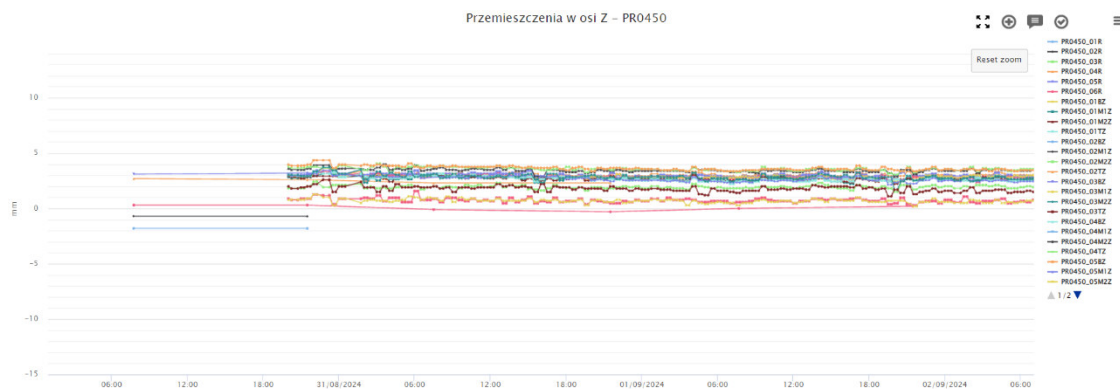


Figure 6-5 Settlement of building PR0450

On PR0460, monitoring points show a downward trend for points PR0460_07R, PR0460_A_13BZ, PR0460_A_13M1Z, PR0460_A_13M2Z and PR0460_A_13TZ, closer to tunnel axis, with a maximum settlement of about 8 mm, higher than the attention threshold.

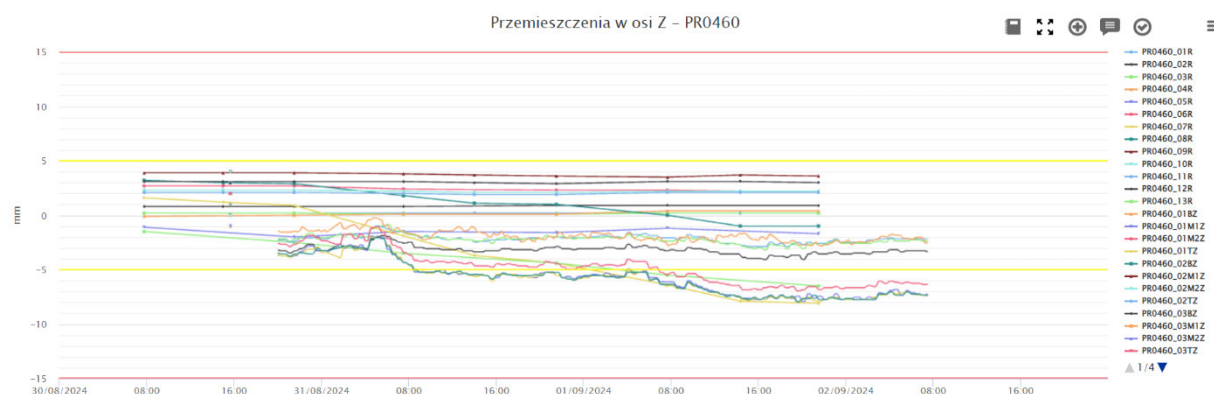


Figure 6-6 Settlement of building PR0460

On MA0230, monitoring points are stable.

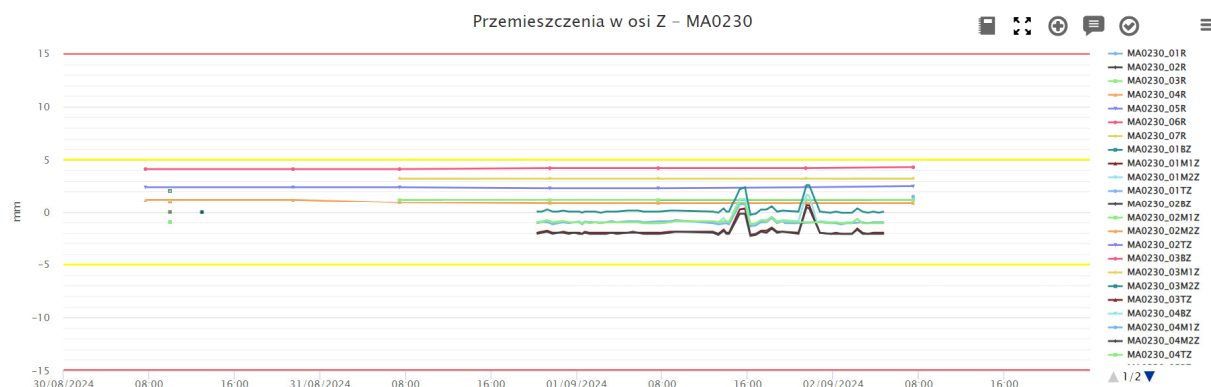


Figure 6-7 Settlement of building MA0230

7. COMPENSATION GROUTING ACTIVITIES

According to Raport Dobowy 30/08/2024 of Soletanche, compensation grouting has been performed on the 30/08/2024 from Shaft 3 in Próchnika 42. The total amount of declared injected volume is 21038.53 l.

According to Raport Dobowy 31/08/2024 of Soletanche, compensation grouting has been performed on the 31/08/2024 from Shaft 3 in Próchnika 42. The total amount of declared injected volume is 46305.74 l.

According to Raport Dobowy 01/09/2024 of Soletanche, compensation grouting has been performed on the 01/09/2024 from Shaft 3 in Próchnika 42. The total amount of declared injected volume is 38502.48 l.

8. SOIL CONDITIONING

The volume of injected foam is plotted in the following Figure. The FIR value is computed based on the excavated material amount per ring and reported in the Table.

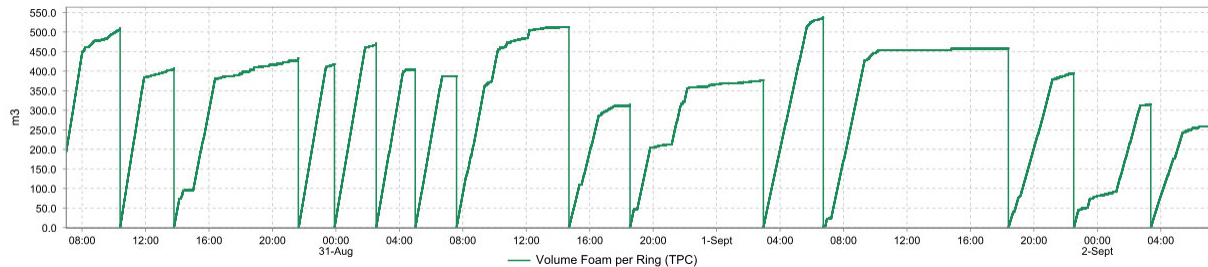


Figure 8-1 Injected foam volume

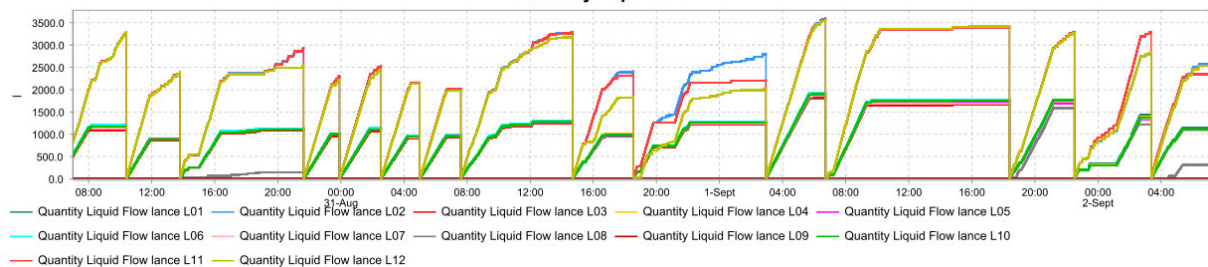


Figure 8-2 Conditioning liquid volume

Table 8-1 Soil conditioning data

Ring	Liquid volume (m³)	C (%)	FIR (%)
238	19.1	2	239
239	14.3	2	191
240	16.4	2	203
241	14.8	2	197
242	16.4	2	221
243	0	0	184
244	13.7	2	182
245	20	2	242

Ring	Liquid volume (m³)	C (%)	FIR (%)
246	14.4	2	148
247	17.1	2	178
248	25.8	2	254
249	24	2	217
250	23.6	1.9	187
251	19.8	1.8	150
252	15.5	2	123